§ 5.5.1

$\boxed{\text{Review}}$ Theorem 5.34

How $f: \mathbb{N}_0 \to \mathbb{N}_0$ can be implemented by a $\underbrace{\text{quantum circuit}}$?

We use a circuit with ancilla (auxiliary register).

Today's goal ; Circuit which implement addition

In appendix B.

$\boxed{\text{Cor B.2}}$ Let $n \in \mathbb{N}$ and $a, b \in \mathbb{N}_0$ with $a, b < 2^n$.

$$a = \sum_{j=0}^{n-1} a_j 2^j \ , \ b = \sum_{j=0}^{n-1} b_j 2^j \ , \ a_j, b_j \in \{0, 1\}$$

We define $c_0^+ := 0$

$$c_j^+ := a_{j-1} b_{j-1} \overset{2}{\oplus} a_{j-1} c_{j-1}^+ \overset{2}{\oplus} b_{j-1} c_{j-1}^+ , \quad 1 \le j \le n$$

$$s_j := a_j \overset{2}{\oplus} b_j \overset{2}{\oplus} c_j^+ , \quad 0 \le j \le n-1$$

Then, $a + b = \sum_{j=0}^{n-1} s_j 2^j + c_n^+ 2^n$

Define operators on ${}^{\P}\mathbb{H}^{\otimes 3}$,

$$A := \mathbb{1}^{\otimes 3} + (X - \mathbb{1}) \otimes |1\rangle\langle 1| \otimes \mathbb{1} ,$$

$$B := \mathbb{1}^{\otimes 3} + (X - \mathbb{1}) \otimes \mathbb{1} \otimes |1\rangle\langle 1| .$$

$$U_s := BA .$$

where X ; Q-Not gate

Note $(|1\rangle\langle 1|)^* = |1\rangle\langle 1|$ and $(|1\rangle\langle 1|)^2 = |1\rangle\langle 1|$

$$X^* = X \text{ and } 2(X - \mathbb{1}) + (X - \mathbb{1})^2 = 0$$

$\Rightarrow A$ is self-adjoint

$$A^2 = \mathbb{1}^{\otimes 3} + 2(X - \mathbb{1}) \otimes |1\rangle\langle 1| \otimes \mathbb{1} + (X - \mathbb{1})^2 \otimes |1\rangle\langle 1| \otimes \mathbb{1} = \mathbb{1}^{\otimes 3}$$

Hence, A is unitary.

In same way, B is unitary

U_S is unitarr.

Since $AB = BA \Rightarrow U_S^* = (BA)^* = AB = BA = U_S$

$$A(|x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |x_1 \overset{2}{\oplus} x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle$$

$$B(|x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |x_0 \overset{2}{\oplus} x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle$$

$$\Rightarrow U_S(|x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |x_0 \overset{2}{\oplus} x_1 \overset{2}{\oplus} x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle$$

$$\Rightarrow U_S(|b_j\rangle \otimes |a_j\rangle \otimes |c_j^+\rangle) = |s_j\rangle \otimes |a_j\rangle \otimes |c_j^+\rangle$$

Define operators on ${}^{q}\mathcal{H}^{\otimes 4}$

unitary
self-adjoint

$$\begin{cases} C := \mathbb{1}^{\otimes 4} + (X - \mathbb{1}) \otimes |1\rangle\langle 1| \otimes |1\rangle\langle 1| \otimes \mathbb{1} \\ D := \mathbb{1}^{\otimes 4} + \mathbb{1} \otimes (X - \mathbb{1}) \otimes |1\rangle\langle 1| \otimes \mathbb{1} \\ E := \mathbb{1}^{\otimes 4} + (X - \mathbb{1}) \otimes |1\rangle\langle 1| \otimes \mathbb{1} \otimes |1\rangle\langle 1| \end{cases}$$

$U_C := EDC$; unitary.

$U_C^* = CDE$

$$C(|x_3\rangle \otimes |x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |x_1 x_2 \overset{2}{\oplus} x_3\rangle \otimes |x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle$$

$$D(|x_3\rangle \otimes |x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |x_3\rangle \otimes |x_1 \overset{2}{\oplus} x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle$$

$$E(|x_3\rangle \otimes |x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |x_0 x_2 \overset{2}{\oplus} x_3\rangle \otimes |x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle$$

$$\Rightarrow \begin{cases} U_C(|x_3\rangle \otimes |x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |x_0(x_1 \overset{2}{\oplus} x_2) \overset{2}{\oplus} x_1 x_2 \overset{2}{\oplus} x_3\rangle \\ \qquad \otimes |x_1 \overset{2}{\oplus} x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle \\ U_C^*(|x_3\rangle \otimes |x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |(x_0 \overset{2}{\oplus} x_1) x_2 \overset{2}{\oplus} x_1 \overset{2}{\oplus} x_3\rangle \\ \qquad \otimes |x_1 \overset{2}{\oplus} x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle \end{cases}$$

$$U_C\left(|0\rangle \otimes |b_{j-1}\rangle \otimes |a_{j-1}\rangle \otimes |c_{j-1}^+\rangle\right)$$

$$= \underbrace{|\underbrace{c_{j-1}^+(a_{j-1} \overset{2}{\oplus} b_{j-1})}_{=c_{j-1}^+ a_{j-1} \overset{2}{\oplus} c_{j-1}^+ b_{j-1}} \overset{2}{\oplus} a_{j-1}b_{j-1}\rangle}_{=|c_j^+\rangle} \otimes |a_{j-1} \overset{2}{\oplus} b_{j-1}\rangle \otimes |a_{j-1}\rangle \otimes |c_{j-1}^+\rangle$$

$\boxed{\text{Def 5.36}}$ $n \in \mathbb{N}$, $\mathbb{H}^B := {}^q\mathbb{H}^{\otimes n+1}$, $\mathbb{H}^A := {}^q\mathbb{H}^{\otimes n}$, $\mathbb{H}^W := {}^q\mathbb{H}^{\otimes n}$

For $|b\rangle \otimes |a\rangle \otimes |w\rangle \in \mathbb{H}^B \otimes \mathbb{H}^A \otimes \mathbb{H}^W$, we define

$$U_0(|b\rangle \otimes |a\rangle \otimes |w\rangle) := |b_n\rangle \otimes \bigotimes_{l=n-1}^{0}\left(|b_l\rangle \otimes |a_l\rangle \otimes |w_l\rangle\right)$$

$$=: |\Psi[b, a, w]\rangle \in \mathbb{H}^B \otimes \mathbb{H}^A \otimes \mathbb{H}^W$$

$$U_1 := \prod_{l=1}^{n-1}\left(\mathbb{1}^{\otimes 3l} \otimes U_C \otimes \mathbb{1}^{\otimes 3(n-1-l)}\right)$$

$$U_2 := \left[(\mathbb{1} \otimes U_S)(\mathbb{1} \otimes \Lambda_{|1\rangle^1}(X) \otimes \mathbb{1})U_C\right] \otimes \mathbb{1}^{\otimes 3(n-1)}$$

$$U_3 := \prod_{l=n-1}^{1}\left(\mathbb{1}^{\otimes 3l} \otimes (\mathbb{1} \otimes U_S)U_C^* \otimes \mathbb{1}^{\otimes 3(n-1-l)}\right)$$

$$\hat{U}_+ := U_0^* U_3 U_2 U_1 U_0$$

$\boxed{\text{Remark}}$ ① In the proof of $\boxed{\text{Thm 5.38}}$, there will be precise explanation of operators.

② $\Lambda_{|1\rangle^1}(X) = \mathbb{1}^{\otimes 2} + (X - \mathbb{1}) \otimes |1\rangle\langle 1|$

$\boxed{\text{Lemma 5.37}}$ U_0, U_1, U_2, U_3 and $\hat{U}_+$ are unitary

pf) ① U_0 is unitary.

U_0 ; change of bases $\Rightarrow U_0$ is unitary.

② U_1, U_3 are unitary

③ U_2 is unitary

$$U_2^* = [U_c^*(\mathbb{1} \otimes \underbrace{\Lambda_{|1\rangle^s}(X)^*}_{= \Lambda_{|1\rangle^s}(X)} \otimes \mathbb{1})(\mathbb{1} \otimes \underbrace{U_s^*}_{= U_s})] \otimes \mathbb{1}^{\otimes 3(n-1)}$$

$$U_2^* U_2 = (U_c^* U_c) \otimes \mathbb{1}^{\otimes 3(n-1)} = \mathbb{1}^{\otimes 3n+1}$$

④ $\hat{U}_+$ is unitary

$\boxed{\text{Thm 5.38}}$ $\exists U_+$; circuit with ancilla on $\mathbb{H}^{I/O} := \mathbb{H}^B \otimes \mathbb{H}^A$

For any $|\Phi\rangle \in \mathbb{H}^{I/O}$

$$\hat{U}_+(|\Phi\rangle \otimes |0\rangle^n) = (U_+|\Phi\rangle) \otimes |0\rangle^n$$

Furthermore, for $a, b < 2^n$,

$$U_3 U_2 U_1 |\Psi[b, a, 0]\rangle = |\Psi[b+a, a, 0]\rangle$$

$\boxed{\text{Rmk}}$ $|\Psi[b, a, 0]\rangle = U_0(|b\rangle \otimes |a\rangle \otimes |0\rangle^n)$

$$\hat{U}_+(|b\rangle \otimes |a\rangle \otimes |0\rangle^n) = U_0^* |\Psi[b+a, a, 0]\rangle$$

$$= |b+a\rangle \otimes |a\rangle \otimes |0\rangle^n$$

On the other hand, $\hat{U}_+(|b\rangle \otimes |a\rangle \otimes |0\rangle^n) = (U_+(|b\rangle \otimes |a\rangle)) \otimes |0\rangle^n$

$$\Rightarrow U_+(|b\rangle \otimes |a\rangle) = |b+a\rangle \otimes |a\rangle.$$